

Non-Stationary (Case by Case)

非平稳: ① Cyclostationary 周期平稳.

$$\exists T, R_X(t, s) = R_X(t+T, s+T)$$

令 $\theta \sim U(0, T)$, θ is independent of $X(t)$

$$Y(t) = X(t+\theta) \text{ is w.s.s.}$$

$$\begin{aligned} \text{Pf: } R_Y(t, s) &= E(Y(t)Y(s)) = E(X(t+\theta)X(s+\theta)) \\ &= E_\theta \left[E_X(X(t+\theta)X(s+\theta) | \theta) \right] \\ &= E_\theta (R_X(t+\theta, s+\theta)) \\ &= \frac{1}{T} \int_0^T R_X(t+\theta, s+\theta) d\theta \end{aligned}$$

$$\begin{aligned} \underline{\underline{\theta' = s+\theta}} \quad & \frac{1}{T} \int_s^{s+T} R_X(t-s+\theta', \theta') d\theta' \\ &= \frac{1}{T} \int_0^T R_X(t-s+\theta', \theta') d\theta' \quad \square \end{aligned}$$

Mean Square Estimation:

$$Y, X \Rightarrow g(X) \longrightarrow Y \Rightarrow \min E(Y - g(X))^2$$

$$\begin{aligned} E(Y - g(X))^2 &= E(Y - E(Y|X) + E(Y|X) - g(X))^2 \\ &= E(Y - E(Y|X))^2 + E(E(Y|X) - g(X))^2 + 2E[(Y - E(Y|X))(E(Y|X) - g(X))] \end{aligned}$$

$$\text{我们断言: } E[(Y - E(Y|X))(E(Y|X) - g(X))] = 0$$

$$\begin{aligned} \text{Pf: } E_X[E_Y((Y - E(Y|X))(E(Y|X) - g(X)) | X)] \\ &= (E(Y|X) - g(X)) E(Y - E(Y|X) | X) \\ &= (E(Y|X) - g(X)) \cdot 0 \\ &= 0 \quad \square \end{aligned}$$

因此 MSE 下的最优解为 $E(Y|X)$.

Pulse Amplitude Modulation (PAM)

$$X(t) = \sum_{k=-\infty}^{+\infty} \alpha_k \phi(t-kT), \quad \alpha_k \text{ 为 information symbol, 假定 } E(\alpha_k \alpha_m) = R_\alpha(k-m)$$

$$\begin{aligned} R_X(t, s) &= E(X(t)X(s)) = E\left[\left(\sum_k \alpha_k \phi(t-kT)\right)\left(\sum_n \alpha_n \phi(s-nT)\right)\right] \\ &= \sum_k \sum_n E(\alpha_k \alpha_n) \phi(t-kT) \phi(s-nT) \\ &= \sum_k \sum_n R_\alpha(k-n) \phi(t-kT) \phi(s-nT) \end{aligned}$$

$$\text{而 } \forall T', R_X(t+T', s+T') \neq R_X(t, s).$$

故 PAM 不宽平稳. 但是 $T' = T$ 时, 满足周期平稳.

$$R_X(t+T, s+T) = R_X(t, s).$$

$$\begin{aligned} \frac{1}{T} \int_0^T R_X(t+\theta, s+\theta) d\theta &= \frac{1}{T} \int_0^T \sum_{k=-\infty}^{+\infty} \sum_{n=-\infty}^{+\infty} R_\alpha(k-n) \phi(t+\theta-kT) \phi(s+\theta-nT) d\theta \\ &= \frac{1}{T} \sum_{k=-\infty}^{+\infty} \sum_{n=-\infty}^{+\infty} R_\alpha(k-n) \int_0^T \phi(t+\theta-kT) \phi(s+\theta-nT) d\theta \end{aligned}$$

$$\text{取 } \theta' = n + \theta - kT, \text{ 则: } \int_0^T \phi(t + \theta - kT) \phi(n + \theta - kT) d\theta \\ = \int_{S-nT}^{S-(n-1)T} \phi(\theta' + t - s - (k-n)T) \phi(\theta') d\theta'$$

取 $k' = k - n$, $n' = n$, 则有:

$$\frac{1}{T} \int_0^T R_X(t + \theta, s + \theta) d\theta = \frac{1}{T} \sum_{k'=-\infty}^{+\infty} \sum_{n'=-\infty}^{+\infty} R_X(k') \int_{S-n'T}^{S-(n'-1)T} \phi(\theta' + t - s - k'T) \phi(\theta') d\theta' \\ = \frac{1}{T} \sum_{k'=-\infty}^{+\infty} R_X(k') \int_{-\infty}^{+\infty} \phi(\theta' + t - s - k'T) \phi(\theta') d\theta'$$

$$\text{引入: } R_\phi(\tau) = \int_{-\infty}^{+\infty} \phi(\theta' + \tau) \phi(\theta') d\theta'$$

$$\text{则 } \frac{1}{T} \int_0^T R_X(t + \theta, s + \theta) d\theta = \frac{1}{T} \sum_{k'=-\infty}^{+\infty} R_X(k') R_\phi(\tau - k'T)$$

$$\frac{1}{T} \int_{-\infty}^{+\infty} \sum_{k'=-\infty}^{+\infty} R_X(k') R_\phi(\tau - k'T) \exp(-j\omega\tau) d\tau \quad (\tau' = \tau - k'T)$$

$$= \frac{1}{T} \sum_{k'=-\infty}^{+\infty} R_X(k') \exp(-j\omega k'T) \int_{-\infty}^{+\infty} R_\phi(\tau') \exp(-j\omega\tau') d\tau'$$

$$= \frac{1}{T} S_X(\omega T) \cdot S_\phi(\omega)$$

② 非平稳: Orthogonal Increment

$$X(t_1), X(t_0) = 0, \quad \forall t_1 < t_2 \leq t_3 \leq t_4, \quad X(t_4) - X(t_3) \perp X(t_3) - X(t_1)$$

$$R_X(t, s) = E(X(t)X(s)) = E(X(t_1) - X(s) + X(s))(X(s) - X(t_0)) \quad , \quad t > s \\ = E[X(s)(X(s) - X(t_0))] \\ = E(X^2(s)) = E[X^2(\min(t, s))]$$

$$\text{Assume } R_X(t, s) = g(\min(s, t))$$

$$E(X(t_4) - X(t_3))(X(t_2) - X(t_1)) \\ = R_X(t_4, t_2) + R_X(t_3, t_1) - R_X(t_3, t_2) - R_X(t_4, t_1) \\ = g(t_2) + g(t_1) - g(t_2) - g(t_1) = 0$$

e.g. Brown Motion

Def: ① $B(0) = 0$ ② Orthogonal Increment ③ $B(t) - B(s) \sim N(0, \sigma^2(t-s))$

$$R_B(t, s) = E[B^2(\min(t, s))] = \text{Var}[B(\min(t, s))] \\ = \sigma^2 \min(t, s)$$

$$Y(t) = \frac{d}{dt} B(t)$$

$$R_Y(t, s) = E(Y(t)Y(s)) = E\left(\frac{d}{dt} B(t) \frac{d}{ds} B(s)\right) \\ = \frac{\partial^2}{\partial t \partial s} E(B(t)B(s)) = \frac{\partial^2}{\partial t \partial s} R_B(t, s) \\ = \frac{\partial^2}{\partial t \partial s} \sigma^2 \min(t, s) \\ = \frac{\partial^2}{\partial t \partial s} \sigma^2 \frac{1}{2} (t + s - |t - s|) \\ = -\frac{1}{2} \sigma^2 \frac{d}{ds} \text{sgn } |t - s| = \sigma^2 \delta(t - s)$$